

Digital Platforms and the Nature of the Firm

Laurent Baronian

Abstract: Digital platforms turn traditional approaches of the firm, which relied on the wage relation to explain the major difference between firm and market, upside down and underline the advantages of coordination through organization over coordination through market. This study aims to propose a definition of the firm able to integrate, besides the integrated firm, also hybrid forms such as networks of subcontractor/subcontracting firms as well as atypical forms such as digital platforms. By reactivating the firm-boundary problem, this article suggests putting valorization by labor at the heart of the firm's decisions concerning integration. It suggests therefore a general definition of the firm as a techno-institutional center of capital valorization, provided that firms make profits by means of the appropriation of labor incorporated into their (productive, structural, intellectual) capital through institutional arrangements. By stressing the relation of production between the owners of the means of production and the direct producers, the approach of the firm supported here should allow to cover the different existing models of the firm, from the classical firm to hybrid models, around which the boundary debate has revolved, to digital platforms.

Keywords: digital platforms, firm theory, boundaries of the firm, transaction cost economics, property rights theory, resource-based theory of the firm, hybrid models

JEL Classification Codes: L14, L24, L25, L62, O30

Digital platforms turn traditional approaches of the firm, which relied upon the wage relation to explain the major difference between firm and market, upside down and underline the advantages of coordination through organization over coordination through market (Bauwens, Mendoza and Iacomella 2012; Botsman 2013; Hamari, Sjöklint and Ukkonen 2015). Yet platform economics does not justify integration, neither by the reduction of transaction costs nor by the creation of organizational resources. Therefore, it calls for a renewal of the firm-boundary problem and the very definition of the firm. By their various modes of appropriation of the producers or consumers' activity, however, digital platforms have put labor at the center of their valorization and therefore of their business models. But these various forms of labor appropriation exceed the framework of the traditional wage relation and reactivate, as we will see, ancient models of exploitation such as the putting-out system. Digital platforms demonstrate what Ayres had formally theorized about the relationship between technological development and institutional structures, namely that Information and communication technologies (ICTs) have allowed a specifically capitalist

Laurent Baronian is associate professor at the University of Sorbonne Paris Nord. He has published articles in history of economic thought, labor economics and money and finance. He published a book entitled Marx and Living Labour (Routledge, 2013) among others. His current research is on knowledge management, financialization of corporate governance, and boundaries of the firm.

form of exploitation without wage labor. Here, technological changes are the sources of institutional changes regarding the labor power (Ayres 1944, 177). What characterizes indeed many of the digital platforms is that they do not need to hire employees for extracting value from their labor. By means of software programs, digital firms make profit instead by inserting themselves “between those who offer services and others who are looking for them, thereby embedding extractive processes into social interactions” (Scholz 2016, 4). Commons defined the corporation as a going concern combining the going plant and the going business (Commons 1924; Guelzo 1976). But digital platforms are going concerns in which the going plant is reduced to some algorithms and software controlling the social activity for the profit of the going business. Most of the transactions are therefore business transactions. Since digital platforms use market relations extensively for valorizing their assets, the firm cannot be distinguished from market regarding transaction costs as supported by transaction-cost economics. Moreover, digital platforms question the resource-based theory of the firm since organizational resources are not located within a firm through management of co-operating hired workers. By subjecting potentially the whole of social activity, digital platforms have completely realized the Veblenian concept of predatory business enterprise (Veblen 1923, 89; Jo 2018). In other words, platform economics paradoxically demonstrate more than traditional firms that the capitalist subjection of labor relies entirely upon the economic power of the owners of the means of the production which are at the source of new forms of cooperation between producers, consumers, users, et cetera, on which its business models rely (Bauwens, Mendoza, and Iacomella 2012).

This article suggests putting valorization by labor at the heart of the firm’s strategies. “By turning his money into commodities that serve as the natural elements of a new product,” says Marx in a draft definition of the firm, “and as factors in the labor process, by incorporating living labor with their dead substance, the capitalist at the same time converts value, i.e. past, materialized, and dead labor into capital, into value with big value, a live monster that is fruitful and multiplies” (Marx [1867] 2010, 205). Today, digital-based firms are capable of extending their field of exploitation well beyond workers’ direct labor, and capitalist firms in general never cease to find new objects of predation. But since all these new objects, even the environment as such, always need the mediation of man’s activity, the concept of labor called upon here includes, in fact, the social activity insofar as it is somehow appropriated in order to be valorized by firms.

By reactivating the firm-boundary problem, the article will show how labor appropriation steers firms’ decisions concerning integration. It will show notably that neither asset specificity nor the creation of specific resources leads to integration. Consequently, the firm cannot be grasped by its governance structure, as supported by transaction-cost economics, nor by its capabilities and resources produced within a circumscribed space of labor, as resource-based theory argues in objection to transaction-cost economics. But the features shared by quasi-firms with traditional firms and digital platforms prompt us to suggest a definition of the capitalist firm in general as a techno-institutional center of capital valorization or, to put it differently, of appropriation of others’ labor. The firm is a technological center since the relation of production relies upon and is carried out by a productive and organizational technology. But it is also an institutional entity, since it appropriates labor incorporated into their (productive, structural, intellectual) capital through institutional arrangements. As Commons put it, the going concern includes rationing transactions describing a relationship between legal superiors and inferiors. In the rationing transaction, the legal superior must

be able to take something from someone else and can be collective in nature as a board of directors apportioning the corporate budget (Guelzo 1976, 42). By stressing the relation of production between the owners of the means of production and the direct producers, the approach of the firm supported here should allow us to cover the different existing models of the firm, from the classical firm to hybrid models, around which the boundary debate has revolved, to digital platforms. As will be seen, all these kinds of firms share the basic feature of extracting surplus labor either directly through work management, or indirectly through contracts and price mechanisms, or through percentages on services provided and by means of software technologies. In order to emphasize this aspect common to the different kinds of firms, I will show how the definitions of the firm by transaction-cost economics (TCE), property-rights theory (PRT), and resource-based theory (RBT) encounter the firm-boundary problem and how this problem gets resolved by putting the exploitation of social labor at the heart of integration strategies of the firm. The critique of these approaches will gradually show that labor haunts them, calling for a re-definition of the firm which finds—in the digital platforms—its clearest illustration.

The article is structured as follows. The next section shows how digital platforms challenge these theories of the firm, by surpassing the disadvantages of the putting-out system and making profits without authority and hierarchy. After having recalled the history of the firm-boundary problem, emphasis will be put on the way it has been dealt with by TCE, then PRT and finally RBT, in accordance with their respective approaches of the firm. The analysis of quasi-firms, in particular the Toyota quasi-integration system, will show that only a definition of the firm that deals with labor as both producing goods and services and creating surplus value appropriated as profits may account for hybrid models while drawing a clear boundary between firm and market. More generally, quasi-integration confirms the priority of productive logic over logics of transaction costs and capabilities and resources. The penultimate section returns to digital platforms and deals with issues emphasized by the approach of the firm supported here. The last section concludes.

Digital Platforms and the Theory of the Firm

The development of information technologies alongside the growth of Web 2.0 has enabled the development of online platforms that promote user-generated content, sharing, and collaboration (Kaplan and Haenlein 2010). Capital has quickly taken advantage of these new modes of exchange through the creation of new kinds of firms called “digital bridge builders” which insert themselves between supply and demand of these services provided by independent workers (Harris and Krueger 2015), thereby “embedding extractive processes into social interactions” (Scholz 2016, 4).

By connecting individuals and communities potentially all around the world, platform-based firms have blurred the distinction between firm and market more deeply than subcontracting and firms’ networks. Moreover, crowdsourcing blurs the boundaries of the firm not only because it uses “collaboration mass weapons” (Tapscott and Williams 2010) extended to the whole of society, but also because local independent workers potentially have access to the global market. This business model has dramatically challenged the conceptual framework of transaction-cost economics. Indeed, digital platforms seem to have overcome all the disadvantages of the putting-out system that led to the creation of manufactures in the seventeenth century (Williamson 1985; Berg 1985, 85). It is well-known that among

the reasons that led the putting-out system to be replaced by manufactures were the lack of control over labor time and intensity, as well as over-quality of the products made by the craft workers (Marglin 1974, 94–95). The ITCs associated with systems of consumer assessment today allow these disadvantages to be overcome by strictly controlling both labor time and quality of the service accomplished by the independent workers. Moreover, digital platforms benefit from what was the main advantage of the putting-out system, namely the collection of information to which their employees had special access (See Benkler 2002, 410–411). Digital platforms thus reduce information asymmetry as well as agents' opportunism (moral hazard). Since transaction costs are much weaker, the platform-based firm does not lead to internalization of transactions (though software programs stand for some form of internalization). Here, the almost exclusive function of the firm is to capture or extract value by reducing market failures through software, facilitating information exchange and increasing competition between independent workers.

Since Adam Smith, it has been the norm to see wage labor as the typically capitalist relationship of employment, with profits, wages, and rents as the typical forms of income resulting from that relationship. Yet the putting-out system depicts a relation of employment where profits of capital result from the extraction of surplus labor, whereas the worker does not sell her labor power for wages but the product of her labor itself. However, the capitalist nature of this relation of production seems so undisputable to historians and economists that all the debate about the transition to the era of manufactures was focused on the reasons pushing the entrepreneurs to hire the labor power in these factories. In a certain way, it is true that transaction-cost economics, and more generally the conception of the firm as a nexus of contracts, made it possible to include the putting-out kind of employment relationship among the firm's contractual relationships. But the disadvantages of the putting-out system have precisely led TCE to ask, "Why do firms exist?" since the putting-out system can be reduced to a market relationship with high transaction costs due to information asymmetry and moral hazard: high buffer inventories, weak leadership in contracting, local responsiveness and system responsiveness respects, high transportation costs (Williamson 1980, 22). Hence, authority and hierarchy as the way to lower transaction costs related to the contractual relation of employment.

Even though authority and hierarchy have almost completely disappeared from the platform employment relationship (Harris and Krueger 2015, 10), the investments made in technologies necessary to connect the stakeholders or users of the platform allow the platform-based firm to make profits on the services provided through contractual devices, in the same way as the merchant-coordinators were making profits on the individual entrepreneur's work.

Therefore, digital platforms show that, against TCE, the firm may rely on coordination through market as well as internal coordination. They thereby confirm contract theory, according to which firm and market differ only according to the degree of contractual commitment between agents. More generally, digital platforms justify the contractual approach of the firm, because, in this case, the link connecting the worker to the firm is exclusively contractual and her productive activity is realized outside any organizational framework. For PRT, only property institutes the manager's hierarchical authority over the worker, namely the residual control rights on the use of her non-human assets. Yet, the holders of these residual control rights have power or authority over human assets, since employees need physical assets to be productive (Hart and Moore 1990). Contract theory

does not deny the importance of cooperation in the running of non-human assets. According to Armen Alchian and Harold Demsetz (1972), authority and control over the labor of firms' employees, and consequently the very existence of the firm, are indeed justified by imperfect information and moral hazard resulting from the inherently cooperative use of productive assets (Alchian and Demsetz 1972, 799). What is neglected, however, is the fact that this cooperation relies in turn upon the social nature of the assets which constitute and explain the manager's authority and hierarchy. Imperfect information and moral hazard are just the consequences of the fact that human assets are employed in order to valorize non-human assets and concern the bilateral contractual relationship between the employer and the employee. In digital platforms, as well as in traditional firms, the manager's power over the worker relies entirely upon the power to manage the means of production necessary to the workers' social and individual reproduction. By founding the authority and incentives on property, PRT not only makes no distinction between firm and market except in accordance to their degree of property but also neglects the fact that the property of non-human assets determines a relation of production exceeding the framework of interpersonal property rights.

In the capitalist firm in general, the property rights ordering the relations between managers and employees are structured not, as PRT claims, by the collective use of the non-human assets as such, but by the social relation of production under which these non-human assets are used and valorized. Since the use and valorization of non-human assets are planned through contractual and organizational arrangements, we suggest defining the firm as a techno-institutional center of valorization (i.e., of extraction of surplus labor). By focusing on the way labor is organized and extracted in order to valorize capital, this approach of the firm may renew and clarify the problem of the firm's boundaries.

The Coasian Genesis of the Firm-Boundary Problem

Ronald Coase published his first article on the nature of the firm in 1937, one year after the publication of Keynes' *General Theory* (1936), which reversed the traditional relation between supply and demand in the equilibrium of prices and quantities. The starting point of Coase's argument is: "why do firms exist, since the market is supposed to ensure the optimal allocation of economic agents' resources?" His question implies that standard theory only dealt with owners of goods, services or assets, and that production could be de jure totally realized through market exchange, as in the putting-out system. Moreover, the theory of factor productivity should demonstrate that labor, land, and capital are paid according to their productivity, provided that they freely exchange on the market (see Clark 1907, 144). Therefore, Coase's question is not "why production?" but rather "why the hierarchical organization of the production?"—in other words—"why the capitalist firm?" Coase agrees with the standard dogma according to which the market is the optimal way to allocate resources in the short run. But standard approaches of the firm have justified the firm only by the necessity to combine factors in such a way that profits be maximized, in other words, that supply and demand for goods and factors be in equilibrium). However, Coase notes that the firm exists in order to coordinate long-term exchange relations which are prone to uncertainty and information asymmetry (Coase 1937, 391). Put differently, the manager's hierarchical authority has to replace prices and quantities in order to coordinate the factors of production, in other words, to equalize the incomes of factors with their productivity

and balance production (Palermo 2000). Coase has thus left an ambivalent contribution to the theory of the firm. On the one hand, he reintegrated within standard theory issues which had been deliberately discarded, such as production, labor management, authority, et cetera. On the other hand, he introduced within the firm the categories specific to market relations of exchange (Pitelis 1993). What will become TCE with Oliver Williamson (1975) is precisely to elaborate on marginal foundations a theory of production, division of labor, capitalist enterprise, et cetera, namely elements which were excluded from economic theory. The new theory should therefore explain and justify on these very foundations the existence of all the excluded economic relations, particularly the capital/labor relationship within the firm. Since the market is the prior institution of the contracting parties, choosing the firm as a means to coordinate assets should only result from market failures (Arrow 1974). Coase's theory of the firm therefore does not deal with the origins of production under capitalism, but directly starts with the market economy whose failures push the producers or asset owners to set up mechanisms of coordination not covered by market coordination: "In the beginning, there were markets," said Williamson (1975, 20). But since the basic assumptions of standard theory are preserved, notably methodological individualism and preference maximization, TCE assumes a vision of the firm primarily as a bundle of bilateral exchange relations, not as a proper institutional space of production organization.

By defining the firm as a mode of coordination alternative to market and as a solution to market failures, Coase's theory is closely related to a specific phase of capitalism during which "The Nature of the Firm" was written. The growth of large corporations, the power of firms to control prices and markets, the scientific management of labor and production within multidivisional firms, et cetera, stress the fact that firms control the markets rather than reacting to their signals (Chandler 1977). Whereas William Lazonick argues correctly that Coase "casts the firm as a passive player that arises out of 'market failure' rather than 'organizational success'" (Lazonick 1991, 169), the point of the rise of the modern firm at that time was rather to offset the weakness of market as a coordinating power by transferring its basic mechanisms to non-market institutions such as corporations. Before being a proper theory of the firm, Coase's theory is, in the first place, a theory of market decline arising from the development of capitalism.

The Fuzzy Boundaries of the Firm in Transaction-Cost Theory

The firm-boundary problem arose again and developed from the 1980s onwards, when liberalization and deregulation of financial markets launched the second wave of mergers and acquisitions, while financial analysts serving institutional investors forced large corporations to fall back on their core competency and outsource numerous parts of the value chain (Prahalad and Hamel 1990). Hence the renewed interest in TCE, which gives theoretical tools to the firm's choice to make or buy and sees the firm and its boundaries in terms of exchange relations, not only between managers and employees, but between the firm and its suppliers, subcontractors, customers, et cetera (Williamson 1975). From this viewpoint, the boundary problem was closely related to the hold-up problem arising as soon as firms make investments in specific assets (Klein, Crawford, and Alchian 1978). The subcontractor investing in specific assets may thus be threatened by an unfavorable renegotiation of contracts by the contracting firm, which may take advantage of the fact that the valorization of the subcontractor's assets depends exclusively on the performance of its

own assets. Conversely, the contracting firm is prone to opportunism for the subcontractor who has the monopoly on the goods or services it provides (Klein, Crawford, and Alchian 1978, 299).

Regarding the formal relationship between economic agents, TCE makes no difference between labor, inter-firm, and market relationships, which are all contractual relationships subjected to information asymmetry and opportunism. Differences concern only the nature of the contract linking the parties within a continuum between the firm and the market. It is true that Williamson sees authority as the distinguishing feature of the relation between the manager and its employees (Williamson 1975, 71–72). But authority may apply to very different forms of organizations. Moreover, Williamson associates the authority relation to the firm because of the employees' character and behavior within the firm, not because of the nature of the firm's functions (i.e., not because of what the firm makes). According to TCE, what distinguishes the firm from the market and justifies it is not the specific nature of the firm's activity (production) but the contractual relationship specific to this activity (the employment relation).

However, the employment relation is only a part of the contractual relationships of the firm. The strength of TCE comes precisely from its ability to conceptualize all of these relationships. According to Williamson, one should not draw any distinction between transactions on the (internal or external) labor market and transactions on the market of intermediary products (Williamson 1986, 87), nor therefore between the assembly of multiple tasks under the manager's authority and the assembly of groups of tasks which are technologically distinct from each other (Williamson 1975, 82). Insofar as transactions are concerned, Williamson sees the same factors at the source of the respective costs and advantages of firms and markets.

Nevertheless, though TCE gives a theoretical framework to integration strategies, the boundaries of the firm seem to remain theoretically ill-defined. Why would savings in transaction costs not lead to the integration of the entire production by one giant firm? In other words, why not a planned economy? Following Coase, Williamson discards this possibility by emphasizing the rise of bureaucratic costs and erosion of incentives (Williamson 1975, 126–130). But insofar as integration choice depends on differences between internal and external transaction costs, what excludes *de jure* the possibility that integration costs would always be weaker than market transaction costs, or, which is the same, that technological efficiency would be always higher than transaction costs and weakening incentives? Moreover, progressive integration of the entire production could favor technological nonseparability to such an extent that it would justify internal control of human and non-human assets (Alchian and Demsetz 1972). This problem emphasizes the failure of TCE to understand the firm as a process of production distinct from exchange relations rather than just in accordance with the market (i.e., only as a solution to market failures). Even though TCE founds the firm on the manager's authority over potentially opportunistic employees and even incorporates dynamic adaptation processes (Williamson 1999), the firm is nonetheless understood as an institution whose choices are exclusively steered by financial strategies. Not only does TCE embrace the manager's point of view by theorizing her choices and trade-offs, but it also grasps her activity as exclusively concerned with the profits of the firm and disconnected from the management of the production process. Differences between TCE and PRT, which I will soon address, has less to do with differences in control procedures against opportunism (Quélin 2002, 65–66) than with the function of firm management. In

PRT, management is required by the aggregation of human and non-human assets in the production process, whereas in TCE, it is required by the aggregation of exchange relations for which hierarchical authority reduces the transaction costs. Since PRT's critique of TCE is based on the requirements of the combination of human and non-human assets necessary to the productive activity and thus to the valorization process, let us look at the way it solves the boundary-problem of the firm.

Property Rights Theory's Critique of Transaction-Cost Theory and its Failure to Solve the Integration Problem

PRT's vision of the firm comes closer to its productive purpose by linking the manager's role to ownership of tradable assets, and thus by seeing the relations within the firm as social relations of production. As defined by Alchian and Demsetz (1972), the firm is a team production in which the combination of assets to produce output requires the control of productivity and distribution of incomes between human assets by a manager emerging from the team. What characterizes the capitalist firm is joint input production with numerous owners, among whom one (the manager) is common to all the contracts of the joint inputs and holds the residual claim as an incentive to monitor the assets. Alchian and Demsetz's firm is like an auction market managed by a kind of Walrasian auctioneer since the manager has no authority over other team members. She has to protect them (Blair and Stout 1999, 253) since her function is to guarantee the optimum use of inputs and the fair income distribution to the factors. Among collective organizations, only the capitalist firm, as a profit-making organization, seems to require a manager for controlling possible shirking by other team members (Alchian and Demsetz 1972, 795). Oliver Hart claims that PRT answers the question that TCE is unable to solve: "Why are the risks of hold-up minimized with integration and what are the costs of integration?" According to Hart, what integration changes are the residual claims, notably concerning the use of assets (Hart 1995). Therefore, he defines the firm with Sanford Grossman as an entity owning means of production (Grossman and Hart 1986, 692) and argues that ownership permits holding all the rights unspecified by the contract (i.e., the residual rights). "Ownership is the purchase of the residual rights to control" (Grossman and Hart 1986, 692).

For Hart and Grossman, integration does not rule out opportunism, as claimed by TCE. Moreover, the assumptions of TCE give no sound reason to limit the size of the firm (Grossman and Hart 1986, 692–693). If we do not dismiss the totally abstract neoclassical assumption of maximization under constraint of decreasing returns to scale (Williamson 1985, 93), trade-off between internal production costs and market transaction costs theoretically sets no limit to the expansion of the firm. According to PRT, integration must rather be understood in terms of ownership and residual rights of control: "integration in itself does not change the cost of writing down a particular contractual provision. What does change is who has control over those provisions not included in the contract" (Grossman and Hart 1986, 695). But integration also entails costs, insofar as ownership transfer reduces the seller's rights and thus her incentives (Grossman and Hart 1986, 693). There is no criterion except the practice allowing to decide whether the incentives to opportunism are higher than the advantages of integration (i.e., residual rights of control), particularly regarding the profits resulting from integration. Moreover, in PRT trade-off between integration and outsourcing depends exclusively on circumstances: since residual rights of control allow a

higher part of income to be derived from inputs, integration would be “optimal when one firm’s investment decision is particularly important relative to the other firm’s.” In addition, nonintegration is desirable when both investment decisions are ‘somewhat’ important” (Grossman and Hart, 716–717). As in TCE, the choice of integration remains theoretically indeterminate.

Although they follow Oliver Hart and John Moore’s property rights vision of the firm, Bengt Holmström and John Roberts have criticized them for underrating other benefits of non-human assets ownership than measurement cost and control, such as incentives and productive efficiency (Holmström and Roberts 1998). They noted that many firms, notably under the subcontracting system, do not decide in favor of vertical integration even though they face very high levels of uncertainty, frequent renegotiations, and use of very specific assets, (see also Dyer 1997, 543–544). Moreover, co-dependent assets and frequent renegotiations seem to favor cooperation between independent firms rather than integration. Indeed, for Holmström the primary reason for integration is not avoiding risks of hold-up, but rather having power over human capital thanks to ownership and control of non-human assets, assigning tasks, delegating authority, and setting the principles that structure implicit and explicit incentives (Holmström 1999, 76). Holmström’s incentive theory indeed claims to provide a genesis of the firm as a bundle of property rights (Holmström 1999). Whereas PRT allows an explanation of how assets are distributed between firms, given a set of assets and investment activities, it fails to explain how these property rights are created. In order to grasp it and thus account for the very existence of the firm and its boundaries, it is more important to explain how and why investment activities are themselves distributed. According to Holmström, the boundary question is what the firms do rather than what they possess (Holmström and Roberts 1998, 87). From this perspective, understanding the structure of *assets* would only amount to understanding the structure of investment *activities*.

Nonetheless, this move from assets to activities brings out new questions: why do firms (or managers) own mostly non-human assets used in production? And conversely, why do workers own hardly any such assets? Put differently, why does the capitalist firm exist, insofar as it is based on the fact that workers do not own their means of production? The ultimate question is therefore: Why property? Why firms, insofar as they are defined by ownership of tradable assets? All the suggestions made by Holmström assume as given precisely what is to be explained, namely the possibility for a firm to possess non-human assets in order to have power over human assets. He suggests that firms own non-human assets, because this reinforces their bargaining power with external parties or because this influences the terms of financing assets. But the benefits of owning assets do not explain how and why these assets may be privately owned. Holmström claims that the most important reason to own non-human assets is to be able to assign human assets to tasks in a way that is more profitable than if ownership were not concentrated. But one of the questions was precisely: why do workers not own non-human assets (Holmström 1999, 88–89)? More generally, although Holmström and Roberts show the benefits of owning means of production and thus of making rather than buying, they do not explain the conditions allowing firms to own these means and take advantage of them by employing human assets deprived of these means. They show how extensive is the managers’ power resulting from ownership of tradable assets, but it is as if efficiency and incentives served as an explanation of the power relation between managers and employees. If opportunism, efficiency, or control certainly explain the firm’s choice to make or buy, they do not account for the very existence of the firm (i.e., ownership

of assets). Moreover, as will be shown, control and incentives over human assets do not allow one to delimit the boundaries of the firm, any more than residual rights of control do. However, by emphasizing the importance of incentives and efficiency of human assets in the ownership of non-human assets, Holmström and Roberts have shifted the focus from assets to activities and thus from the firm's structure of property to its joint process of production and valorization.

The Limitless Boundaries of the Resource-Based Firm

In fact, drawing the boundaries of the capitalist firm requires knowing its nature and function. If one understands capital as a mode of production and not just as a source of revenue, as Thomas Piketty (2014) contends, the function of the firm appears as dual and consists in producing goods and services in order to valorize productive investment (i.e., to make profits from investment). What indeed characterizes a capitalist enterprise is that production of goods and services is jointly a valorization process (Marx [1867] 2010, 208). The firm is then a techno-institutional center in which production and distribution of goods and services by labor are the means of valorizing the assets used in that process.

Prima facie, RBT seems to be the only firm model to explicitly link valorization to the production of specific goods or services by defining the firm as a bundle of capabilities, organizational routines, and knowledge produced thanks to a combination of resources and cooperation between stakeholders of the firm (Coriat and Weinstein 2010, 60–62). All these routines, forms of knowledge, and capabilities give the firm a competitive advantage (Barney 1991) resulting in “rents to costly-to-copy productive assets” (Conner 1991, 144). As dynamic capabilities, now firms exist not because of market failures as in TCE, but because markets for planned innovating products do not yet exist (Teece 2010, 175). It is up to the firms to create them according to an enlarged vision of the firm that not only creates value but captures value on the new markets it creates at the same time (Pitelis and Teece 2009, 6). Therefore, firms exist in order to create or co-create the markets necessary to capture value, in addition to creating value, thanks to their competitive advantage and their dynamic capabilities (Pitelis and Teece 2009, 11; see also Teece 2006). Whereas according to Coase and Williamson, the firm is a kind of private market, for Christos Pitelis and David Teece the market appears as a public firm since according to this approach, the basic function of the firm is to create a permanent disequilibrium by means of innovation. Undoubtedly the firm's purpose of profit maximization is reached through innovations and dynamic capabilities that “escape the zero-profit condition” (Augier and Teece 2008, 1190). But this purpose does not relate to the firm as a productive organization. It only demonstrates that production of goods and services is subjected to profit maximization. More generally, the concepts of “capability” or “routine” concern parts of the firm's activity that do not enter the valorization of productive capital but are relied upon in the firm's connections with external parties (customers, suppliers, et cetera). This “relational capital” is certainly a part of the immaterial capital taken into account by financial analysts in the market valuation of the firm (Abhayawansa and Guthrie 2010), but it does not contribute to the valorization of the firm's productive assets.

Furthermore, although Pitelis and Teece claim to be able to grasp the positive nature of the firm in opposition to TCE, which justifies it only negatively as a solution to market failures, their approach does not solve the firm-boundary problem, insofar as it extends to

all the relations of the firm, and thus to the entire economy, the power of creating the value captured by the firm (Pitelis and Teece 2009, 10). How can one delimit the boundaries of a firm whose management is based on communities of users or sharers connected to worldwide communication networks such as digital platforms? The resource-based theory, by founding the identity of the firm on its organizational and collective nature, is prone to include the entire “knowledge society” (or the sharing society) as a stakeholder of the firm. For different reasons, neither TCE and PRT, nor RBT are able to strictly delimit the boundaries of the firm: TCE and RPT because the firm differs from market exclusively with respect to the nature and amount of costs resulting from the incomplete contracts; RBT because the firm is not distinguished from the whole of its connections with all its stakeholders.

But even when the concept of capability is strictly limited to the organization of production, the boundaries of the firm remain undefined. According to RBT, the choice to integrate one or some parts of the value chain depends on the possibility to develop a specific competence (Argyres and Zenger 2012). But as will be analyzed below, specific skills may be developed through hybrid forms of organization in which the firm relies on the market for creating its organizational advantage. Moreover, platform-based firms may rely exclusively on the market for taking advantage of the distributed skills and competences and spontaneous co-operation between users and sharers. As Laura Poppo and Todd Zenger (1998) noted after Jeffrey Bradach and Robert Eccles (1989), “the same exchange conditions such as asset specificity or measurement accuracy may indeed hinder the performance of hierarchical exchanges as well as market performance” (Poppo and Zenger 1998, 854). Moreover, the indeterminate boundaries of the resource-based firm seem to reinforce the position of TCE regarding this issue. Indeed, Nicolai Foss has remarked that “the boundaries of the firm are difficult to rationalize except on a standard contractual basis” (Foss 1996, 475) when firms have the opportunity to communicate knowledge to supplier firms (Dyer 1997, 546). Only this contractual basis apparently justifies the different strategic choices of the firm in order to develop its capabilities by applying to them the criterion of comparison between internal organization costs and market costs (Langlois 1992, 110; Baudry and Chassagnon 2010, 496).

If the firm is defined as a bundle of organizational capabilities, therefore, its boundaries would be delimited by all the assets contributing both to the production of routines and competences and to the process of valorization. It would be therefore necessary to exclude from the boundaries of the firm the suppliers, subcontractors, customers, et cetera who do not take part in the valorization process of productive assets; indeed the economic operations between them and the firm consist in exchange of money for goods or services, notwithstanding information exchange contributing to the creation of the contracting firm’s specific resources. Although the subcontracted output is produced by means of specific assets, it is bought by the contracting firm. Since it is a mere act of exchange, it does not come into the production process. Thus, the concept of capability or resource is either too narrow or too broad for drawing the boundaries of the firm: too narrow, since it excludes the valorization process from the production process; too broad, since some resources are created through relations of exchange with parties external to the contracting firm and are not part of its own valorization process (Langlois 2003). It fuses production relations and market relations under the same concept, even though a clear distinction between them is necessary to delimit the boundaries of the firm within which the valorization process takes place.

The Quasi-Firm-Boundary Problem

We have seen how TCE, PRT, and RBT meet difficulties in dealing with vertical integration and firm-boundary problems. However, these approaches disclose their limitations mostly when they tackle hybrid models. Hybrid models have been analyzed by many scholars, but mostly in order to stress their specific advantages with respect to vertical integration and market relations. Williamson (1991) himself notably insists on close contractual relations as a way to compensate the weak coordination of market relations between the contracting firms (1991, 280). According to Anoop Madhok (2002) however, the benefits of hybrid models surpass transaction costs and allow firms to create and operate resources and capabilities. In addition to governance structure and transaction costs, the characteristics of resources determine the boundary choice of a firm (Madhok 2002, 541). To the extent that interfirm collaboration permits joint production of resources, the distinction between production and exchange is blurred since firms choose hybrid models in order to not only lower their transaction costs but to increase efficiency and create more value. However, Madhok assumes a narrow definition of the quasi-firm as the product of alliance between two or more independent firms. But since the distinction between production and exchange is blurred, how can we delimit the boundaries of the firm? As soon as firms form alliances, “there is a *coevolution* between the competencies of firms and those of their partners” (Madhok 2002, 546). Here again, the quasi-firm issue is dealt with in terms of benefits for firms to adopt hybrid forms, yet by obfuscating the boundary problem arising with these forms. Michael Jacobides and William Billinger show through an empirical study that the boundary problem does not arise from the distinction between buying or making, but that firms may choose mixed and dynamic forms between them (Jacobides and Billinger 2006, 258). Even though their study shows how a firm is led to change its boundaries over time according to valorization strategies, it does not deal with the question: what are the boundaries of a firm engaging in hybrid relations with other firms? The existing literature stresses the advantages of hybrid models which surpass the transaction cost problems and the fact that efficiency and productivity prevail among these advantages (Powell, Koput, and Smith-Doerr 1996; Grant 1996; Dyer 2000). But it does not deal with the boundary problem arising from these hybrid forms in which the distinction of production and exchange is blurred. Interestingly, Mark Granovetter (1985) sees the quasi-firm as closer to vertical integration than to market relations but understands it as interactions between a contractor and subcontractors geographically close to each other where the latter are supervised by the former (Granovetter 1985, 497–498). More generally, he stresses the importance of social and personal relations in the choice of vertical integration or market relations. In the wake of TCE indeed, close social relations between firms may compensate high transaction costs arising from market relations between firms and thus prevent them from vertical integration (1985, 498).

I claimed above that, since subcontracting is a market relation between two or more legally independent firms, it should exclude subcontractors from the boundaries of the contracting firm. This seems to meet the vision of TCE according to which subcontracting is a hybrid market transaction. However, what justifies exclusion of subcontractors from the boundaries of the firm is not the recourse to the market as such, but the fact that the value exchanged between two independent firms does not produce more value than previously created by these firms. Goods or services bought by the contracting firm are certainly a part of the final output, but they have just valorized the productive capital of the subcontractor.

The contracting firm has hypothetically bought them at their value, thus realizing the surplus value (profits) produced by the subcontractor. Things are different when it appropriates a part of the surplus value produced by the subcontractor through transfer pricing mechanisms. In this case, not only is the output part of the final output, but the surplus value incorporated into the former has valorized the contracting firm's assets. Here, everything seems as if market transaction was realized between divisions of the firm through transfer pricing. This case of "hold-up" by the contracting firm is enabled by the specificity of the subcontractor's assets but leads neither to subsequent underinvestment nor to vertical or lateral integration. However, the subcontractor, despite its legal independence, must be seen as included within the boundaries of the contracting firm. This is indeed a case of quasi-integration (Houssiaux 1957, 222¹) which shows the inadequacy of the TCE conception of the firm. Indeed, in a great number of industrial strategies, recourse to market is merely formal and does not relate to incentives or transaction cost problems, but only to productive efficiency or combination and organization of the value chain. On the other hand, quasi-integration confirms that concepts of capability or resource, insofar as they are sources of rents, no longer allow us to delimit the boundaries of the firm. Here, the firm's strategy consists simultaneously of concentrating on its core competencies and appropriating the surplus value produced in the outsourced segments of the value chain. The specific competence of the firm is neither sufficient to define the nature of the firm nor to delimit its boundaries. Quasi-firms show in particular that asset specificity does not necessarily lead to vertical integration. In some cases, it allows the contracting firm to gain power over the subcontractors through transfer pricing by which it appropriates a part of the profits or rents created by subcontractors thanks to their specialized know-how.

Let us take the famous and often-commented case of the Toyota system (Monden 1994; Dyer 2000). The firm uses the "parallel order" system in order to stimulate competition between its suppliers, not only over quality, but also over price (the visible hand of competition; see Itami 1988). In fact, the only flexibility of the suppliers with respect to Toyota is to develop their R&D in order to rationalize production and gain additional margins on their selling prices, which are fixed according to the former production methods. Yet these improvements are also profitable to Toyota (Shimizu 1999, 73–74). Benjamin Coriat has noted that "In Japan—as anywhere else in the world—relations between large and small subcontracting firms are marked by a certain asymmetry showing more or less strongly inequality of economic relations of power between contracting entities" (Coriat 1991, 109; see also Houssiaux 1957, 232). This leads to less investment in fixed capital by the contracting firm, rents on the wage difference between employees of large and small firms, and amortization of cyclical fluctuations effects (Coriat 1991, 109). In fact, the kind of subcontracting specific to Toyota allows the firm to take advantage of all the benefits of integration through contractual relations of market. Furthermore, subcontractors of the Toyota system accumulate specific know-how thanks to their long relationship with the contracting firm, so that this relationship allows continuous inter-firm organizational innovation. This accumulated immaterial capital explains why the contracting firm has difficulty changing its subcontractors, since it would destroy the resources of accumulated know-how (Dyer 2000, 549–550).

The Toyota system illustrates a case where the firm is grasped, beyond its traditional boundaries, by its specific competencies. But it is only insofar as these competencies concern

¹ Jacques Houssiaux is a French industrial economist who is largely unknown even in France, though he is maybe the first to have theorized the concept of quasi-integration.

the process of production, and thus of valorization, that RBT can delimit the boundaries of the Japanese firm more accurately than TCE or PRT. In other words, Toyota's subcontractors have developed not only technical know-how, but organizational know-how, allowing the highest continuity, flexibility and speed for the lean production process (Coriat 1991, 226). RBT therefore not only identifies the firm differently—despite having similar empirical results as TCE (Carter and Hodgson 2006)—but, as we have seen, can indefinitely extend the boundaries of the firm as long as relational capital is not related to the processes of production and valorization of the firm's productive assets. What allows us to distinguish the part of the relational capital relied upon in the valorization process from other parts is that first, it gives rise to relational rents and second, it may be shared between the firm and its subcontractors in theory, something which can never be the case between the firm and its customers. The Japanese firm and its suppliers and subcontractors, as integrated from the perspective of their organizational and technical competencies on the one hand, and contractualized from the perspective of their formal relations on the other, are indeed quasi-integrated and together draw the boundaries of a single firm.

In their seminal article on the boundaries of the firm, Klein, Crawford, and Alchian claim that hold-up threatens not only the supplier or subcontractor which has invested in specific assets, but also the contracting firm, since the subcontractor may hold a monopolistic position vis-à-vis the firm (Klein, Crawford, and Alchian 1978, 299). Theoretically, then, specific assets may threaten the two contracting parties. But how, practically, could a subcontractor, whose assets are totally dedicated to the production of output exclusively bought by the contracting firm, ever exert any pressure on the latter? According to the authors, the threat of hold-up risks bringing about underinvestment—or even no investment at all—from the subcontractor. We may object that the possibility of underinvestment or no investment is a risk to society's well-being and so does not just concern the firm itself, which in any case always takes decisions motivated by profit, though under bounded rationality. Moreover, referring to underinvestment means replacing the framework of positive analysis by normative arguments. The problem at hand is to analyze what happens when a subcontracting relationship occurs, not to state what is likely not to occur. Yet, within this kind of relationship, it seems clear that the contracting firm has power over its subcontractor (Dyer 2000, 358). Moreover, this kind of relationship implies appropriation of the whole or part of the quasi-rents generated by specific investments of the subcontractor. This case does not lead to vertical integration, neither from the contracting firm whose transaction costs are compensated by the amount of the appropriated rent, nor from the supplier whose position gives it no recourse to this option. Its only possibility would be to exit the market. But subcontracting is a strategy decided by the contracting firm itself and most often does not relate to any strategic logic between two or more independent firms. Therefore, it does not come under the logic of transaction costs, which is entirely based on bounded rationality and opportunism. It basically comes under the logic of the valorization allowed by new technological possibilities (NTICs, polyvalent machinery, modular production, et cetera).²

² The fruitfulness which TCE has found in hybrid models comes exclusively from the fact that they use formal mechanisms of market exchange within one and only one global production process (i.e., within a single value chain). The empirical examples that Williamson cites to illustrate the scope of TCE shows indeed how the productive logic subordinates the logic of transaction costs in the inter- and intra-firm relationships. Williamson himself admits that most of vertical integration downstream of distribution falls within technical requirements and know-how: "Integration into final sales and service is mainly observed for consumer and producer durables where considerable knowledge is imparted at point of sale and specialized follow-on service is required" (Williamson 1985, 114). Notwithstanding the doubts cast on the Fisher Body-GM case study (Coase 2006), the fact that GM

Hybrid models such as quasi-integration challenge traditional theories of the firm. Moreover, they show how the joint logic of production and valorization processes explains the nature and stakes of the relationship between contracting firms and delimits clearly their boundaries which remain otherwise blurred. Although contractual relations apparently prevail among the multiple stakeholders, hybrid models show paradoxically the importance of living labor exploitation in the understanding of industrial strategy of the firms. But above all, they show us how digital platforms go one step further in this logic of social activity exploitation.

The Digital Platform-Boundary Solution

We have seen how TCE, PRT, and RBT have faced difficulties in solving the (quasi)firm-boundary problem. TCE—and from this point of view also PRT—clearly distinguishes the firm from the market, but only from the perspective of their contractual relations, not with respect to the specific function of the firm regarding the market: production of goods or services incorporating surplus value appropriated as profits. Consequently, the trade-off between internal organization costs and market costs remains practically indeterminate. Thus, concerning organizational change able to reduce the internal transaction costs of the firm, Williamson unjustifiably asserts that “the qualitative implications” of the argument according to which “the distinctive powers of internal organization are impaired and transactional diseconomies are incurred as firm size and the degree of vertical integration are progressively extended” “are not upset” (Williamson 1975, 117). But nowhere is it explained why they are not upset. Conversely, RBT understands the firm as an institution creating routines and knowledge dedicated to specific goods or services, yet confuses production of surplus value with creation of competitive advantage (which is supposed to be the source of rents or quasi-rents). Thus, according to this approach of the firm as a bundle of resources giving rise to organizational routines or specific knowledge, rents and quasi-rents arise as much from advantages drawn from market relations with its customers, suppliers, et cetera as from efficiency of the production process or specificity of the product. Strangely, each of these approaches fails to draw the boundaries of the firm clearly, being unable to distinguish adequately the firm from the market: TCE and PRT, because they bring the contractual nature of the specific market relation into the firm, even though PRT in particular admits the

bought Fisher Body in 1926 because it did not meet the new requirements of the firm—such as the proximity of the site—leads Williamson to a similar conclusion: “To be sure, transaction cost reasoning does not predict this sequence in detail” (Williamson 1985, 115). About Gordon Walker and David Weber who did a study on Ford and GM suppliers: “Although they interpret their results as ‘mixed’ in transaction cost terms and attribute most of the explanatory power to ‘comparative production costs,’ they acknowledged that this may be due to ‘limitations in both the data and the model for which subsequent work is indicated’” (Williamson 1985, 116). And he adds: “Further effort to plumb the factors that are responsible for comparative production cost differences between buyer and supplier will, I submit, disclose added transaction cost features” (1985, 116). Concerning the integration of a bauxite refinery by steel enterprises, Williamson here again admits that it was decided “because of asset specificity of both physical plant and site specificity kinds,” and that it was “reinforced by information asymmetry considerations” (Williamson 1985, 119). Furthermore, Williamson cites cases of integration that ought to be considered as mistakes from the viewpoint of transaction costs. Without any supporting evidence, he “conjecture[s]” that they “have mainly been abandoned as renewal decisions have presented themselves” (Williamson 1985, 119). It is true that, in the meantime, numerous empirical studies seem to have confirmed Williamson’s assumptions concerning the trade-off between making and buying (Masten 1996; Shelanski and Klein 1995). However, in an article gathering together most of the case studies illustrating “the empirical success story of the TCE” (Williamson 2000, 607), the authors acknowledge that the quasi-firm studied by Eccles (1981) seems to be an exception to the rule that integration choice increases with asset specificity and conclude that “TCE may assign too much attention to asset specificity for the integration decision” (Shelanski and Klein 1995, 346).

co-operative nature of production; and RBT because it does not draw any clear distinction between the advantages of the firm drawn from its market relations and those resulting from the organization of production.

By blurring the distinction between firm and market more than ever before, the business model of digital platforms has practically refuted the theoretical framework of TCE. Moreover, by overcoming the disadvantages of the putting-out system, digital platforms have shaken the two pillars underpinning the TCE approach and its vindication of the capitalist firm, namely information asymmetry and moral hazard (Williamson 1996). But digital platforms have also challenged the PRT according to which authority and hierarchy are the consequences of the residual control of non-human assets (Hart and Moore 1990). As we have seen, digital platforms do not need to exercise direct authority over workers and may control them through anonymous devices (software, consumer assessments, algorithms, et cetera). But if digital platforms may relax direct authority and control, they nevertheless share with all other kinds of capitalist firms the relation of production based on the property and use of non-human assets by capitalists who extract living labor in order to valorize these social means of production. Digital platforms have undermined employment relations based on wages but at the same time they have radicalized the relation of production on which is based wage labor, namely the private ownership of social and collective means of production. This relation of production suggests seeing the firm as a techno-institutional center of extraction of surplus labor. The focus on the private economic power on social production allows us to distinguish effectively between firm and market whatever is the form of hybridization between them. Moreover, it provides the necessary framework to account for different forms of authority and control by relating them to the different ways of extracting living labor and valorizing non-human assets.

Concluding Remarks

Since integration does not delimit the boundaries of the firm, solving the firm-boundary problem depends on a clear understanding of the nature and function of the firm. If we assume that labor is the source of the profits made by the owners of the firm's assets, all the features put to the fore by traditional theories (authority over employees in TCE, control of the use of firm's assets by PRT, management of competencies, and knowledge by RBT) define the firm as a techno-institutional center of labor appropriation, regardless of whether or not such appropriation is directly organized by the firm. Defined as such, the capitalist firm includes digital platforms, where production of value is fragmented into a large number of individual provisions of services, since it frees the firm-boundary problem from governance constraints on transactions as well as organizational constraints on knowledge and resources, in the same way as ICTs have practically freed the firm from those same constraints. One may object that digital economics raises more transaction problems than any other industrial sectors. But the reduction of transaction costs made possible by ICTs does not justify anymore the hierarchical institution of the firm asserted by TCE. Moreover, what distinguishes digital platforms is that they cannot choose between integration and market, since their hybrid form derives from the very nature of the services they produce. In the same way, creation of organizational knowledge in digital platforms relies precisely on the decentralized and scattered nature of knowledge production by the many stakeholders of the digital platform. Thus, organizational resources are not limited to the resources created by the firm's employees

but now extend to all the data produced by the workers and consumers and appropriated by the firm's software which interposes them. But whatever the reasons pushing firms to choose their forms of hybridization or their modes of organizational knowledge, they are always conditioned by the pursuit of the primary objective of the firm, namely the extraction of surplus labor or the valorization of social activity. Since digital platforms make profits much like other capitalist firms by incorporating living labor to non-human assets, the typology of the firms and their organizational modes specific to contemporary capitalism could and should be made in accordance with the different ways in which the connection between non-human and human assets is made.

References

- Abhayawansa, Subhash, and James Guthrie. 2010. "Intellectual Capital and the Capital Market: A Review and Synthesis." *Journal of Human Resource Costing and Accounting* 14 (3): 196–226.
- Alchian, Armen A., and Harold Demsetz. 1972. "Production, Information Costs, and Economic Organization." *American Economic Review* 62 (5): 777–795.
- Argyres, Nicholas S., and Todd R. Zenger. 2012. "Capabilities, Transaction Costs, and Firm Boundaries: A Dynamic Perspective and Integration." *Organization Science* 23 (6): 1643–1657.
- Arrow, Kenneth. 1974. *The Limits of Organization*. New York: W. W. Norton.
- Augier, Mie, and David J. Teece. 2008. "Strategy as Evolution with Design: The Foundations of Dynamic Capabilities and the Role of Managers in the Economic System." *Organization Studies* 29 (08–09): 1187–1208.
- Ayres, Clarence E. 1944. *The Theory of Economic Progress*. Chapel Hill: The University of North Carolina Press.
- Barney, Jay. 1991. "Firm Resources and Sustained Competitive Advantage." *Journal of Management* 17 (1): 99–120.
- Baudry, Bernard, and Virgile Chassagnon. 2010. "The Close Relation between Organization Theory and Oliver Williamson's Transaction Cost Economics: A Theory of the Firm Perspective." *Journal of Institutional Economics* 6 (4): 477–503.
- Bauwens, Michel, Nicolas Mendoza, and Franco Iacomella. 2012. *Synthetic Overview of the Collaborative Economy*. P2P Foundation. Available at: <https://p2pfoundation.net/wp-content/uploads/2018/02/Synthetic-overview-of-the-collaborative-economy.pdf> Accessed on October 30, 2019.
- Benkler, Yochai. 2002. "Coase's Penguin, or Linux and the Nature of the Firm." In *Yale Law Journal* 112 (3): 369–446.
- Berg, Maxine. 1985. *The Age of Manufactures: 1700–1820*. London: Fontana Press.
- Blair, Margaret M., and Lynn A. Stout 1999. "A Team Production Theory of Corporate Law." *Virginia Law Review* 85 (2): 247–328.
- Botsman, Rachel. 2013. *The Sharing Economy Lacks a Shared Definition*. Fast Company. Retrieved from <https://www.fastcompany.com/3022028/the-sharing-economy-lacks-a-shared-definition>. Accessed on December 6, 2019.
- Bradach, Jeffrey L., and Robert G. Eccles. 1989. "Price, Authority, and Trust: From Ideal Types to Plural Forms." *Annual Review of Sociology* 15: 97–118.
- Carter, Richard, and Geoffrey M. Hodgson. 2006. "The Impact of Tests of Transaction Cost Economics on the Debate on the Nature of the Firm." *Strategic Management Journal* 27: 461–476.
- Chandler, Alfred D. 1977. *The Visible Hand. The Managerial Revolution in American Business*. Harvard: Harvard University Press.
- Clark, John B. 1907. *Essentials of Economic Theory as Applied to Modern Problems of Industry and Public Policy*. New York: The Macmillan Company.
- Coase, Ronald H. 1937 "The Nature of the Firm." *Economica* 4 (16): 386–405.
- Coase, Ronald H. 2006. "The Conduct of Economics: The Example of Fisher Body and General Motors." *Journal of Economics and Management Strategy* 15 (2): 255–278.
- Commons, John R. 1924. *Legal Foundations of Capitalism*. New York: The Macmillan Company.
- Conner, Kathleen R. 1991. "A Historical Comparison of Resource-Based Theory and Five Schools of Thought within Industrial Organization Economics: Do We have a New Theory of the Firm?" *Journal of Management* 17 (1): 121–154.

- Coriat, Benjamin. 1991. *Penser à l'envers*. Paris: Christian Bourgois.
- Coriat, Benjamin., and Olivier Weinstein. 2010. "Les théories de la firme entre 'contrats' et 'compétences'. Une revue critique des développements contemporains." *Revue d'Économie Industrielle* 129-130: 57-86.
- Dyer, Jeff H. 1997. "Effective Interim Collaboration: How Firms Minimize Transaction Costs and Maximise Transaction Value." *Strategic Management Journal* 18: 535-556.
- Dyer, Jeff H. 2000. *Collaborative Advantage: Winning Through Extended Enterprise Supplier Networks*. New York: Oxford University Press.
- Eccles, Robert G. 1981. "The Quasifirm in the Construction Industry." *Journal of Economic Behavior and Organization* 2: 335-357.
- Foss, Nicolai J. 1996. "Knowledge-Based Approaches to the Theory of the Firm: Some Critical Comments." *Organization Science* 7 (5): 470-476.
- Granovetter, Mark. 1985. "Economic Action and Social Structure: the Problem of Embeddedness." *American Journal of Sociology* 91 (3): 481-510.
- Grant, R. M. 1996. "Toward a Knowledge-Based Theory of the Firm." *Strategic Management Journal* 17: 109-122.
- Grossman, Sanford J., and Oliver D. Hart. 1986. "The Costs and Benefits of Ownership: A Theory of Lateral and Vertical Integration." *Journal of Political Economy* 94: 691-719.
- Guelzo, Carl M. 1976. "J. R. Commons and the Theory of the Firm." *The American Economist* 20 (2): 40-46.
- Hamari, Juh, Mimmi Sjöklint, and Antti Ukkone. 2016. "The Sharing Economy: Why People Participate in Collaborative Consumption." *Journal of Association for the Information Science and Technology* 67: 2047-2059.
- Hart, Oliver D. 1995. *Firms, Contracts, and Financial Structure*. Oxford: Clarendon Press.
- Hart, Oliver D., and John Moore. 1990. "Property Rights and the Nature of the Firm." *Journal of Political Economy* 98 (6): 1119-1158.
- Harris, Seth D., and Alan B. Krueger. 2015. "A Proposal for Modernizing Labor Laws for Twenty First Century Work: The 'Independent Worker.'" *The Hamilton Project*. Discussion Paper 2015 10.
- Holmström, Bengt, and John Roberts. 1998. "The Boundaries of the Firm Revisited." *Journal of Economic Perspectives* 12 (4): 73-94.
- Holmström, Bengt. 1999. "The Firm as a Subeconomy." *Journal of Law and Economic Behaviour* 15 (1): 74-102.
- Houssiaux, Jacques. 1957. "Le concept de 'quasi-intégration' et le rôle des sous-traitants dans l'industrie." *Revue Économique* 8 (12): 221-247.
- Itami, Hirojuki. 1988. "Competition by Visible Hands: Efficiency of Parts Supply Systems." In *Competition and Innovation; Corporate Development in the Automobile Industry*, edited by Hirojuki Itami, T. Kagono, T. Kobayashi, K. Sakakibara, and M. Ito. Tokyo: Toyo Keizai Shimpō-sha.
- Jacobides, Michael G., and Stephan Billinger. 2006. "Designing the Boundaries of the Firm: From Make, Buy, or Ally to the Dynamic Benefits of Vertical Architecture." *Organization Science* 17 (2): 249-261.
- Jo, Tae-Hee. 2018. "The Institutional Theory of the Business Enterprise: Past, Present, and Future" MPRA Paper No. 84036. Available at <https://mpra.ub.uni-muenchen.de/84036/>, accessed on October 30, 2019.
- Kaplan Andreas M., and Michael Haenlein. 2010. "Users of the World, Unite! The Challenges and Opportunities of Social Media." *Business Horizons* 53 (1): 59-68.
- Keynes, John Maynard. 1936. *General Theory of Employment, Interest and Money*. London: Macmillan.
- Klein, Benjamin, Robert G. Crawford, and Armen A. Alchian. 1978. "Vertical Integration, Appropriable Rents, and the Competitive Contracting Process." *Journal of Law and Economics* 21 (2): 297-326.
- Langlois, Richard N. 1992. "Transaction-Cost Economics in Real Time." *Industrial and Corporate Change* 1 (1): 99-127.
- Langlois, Richard N. 2003. "The Vanishing Hand: The Changing Dynamics of Industrial Capitalism." *Industrial and Corporate Change* 12 (2): 351-385.
- Lazonick, William. 1991. *Business Organization and the Myth of the Market Economy*. New York: Cambridge University Press.
- Madhok, Anoop. 2002. "Reassessing the Fundamentals and Beyond: Ronald Coase, the Transaction Cost and Resource-Based Theories of the Firm and the Institutional Structure of Production." *Strategic Management Journal* 23: 535-550.
- Marglin, Stephen. 1974. "What Do Bosses Do? The Origins and Functions of Hierarchy in Capitalist Production, Part I." *The Review of Radical Political Economics* 6 (2): 60-112.
- Marx, Karl. (1867) 2010. *Capital, Volume I*. New York: Lawrence and Wishart.
- Masten, Scott E. 1996. *Case Studies in Contracting and Organization*. New York: Oxford University Press.
- Monden, Yasuhiro. 1994. *Toyota Production System: An Integrated Approach to Just-In-Time*. London: Chapman and Hall.

- Palermo, Giulio. 2000. "Economic Power and the Firm in New Institutional Economics: Two Conflicting Problems." *Journal of Economic Issues* 34 (3): 573–601.
- Piketty, Thomas. 2014. *Capital in the 21st Century*. Boston: Harvard University Press.
- Pitelis, Chiristos. 1993. "On Transaction (Costs) and Markets and (as) Hierarchies." In *Transaction Costs, Markets, and Hierarchies*, edited by Chiristos Pitelis, 1262–276. Oxford: Basil Blackwell.
- Pitelis, Chiristos, and David Teece. 2009. "The (New) Nature and Essence of the Firm." *European Management Review* 6 (1): 5–15.
- Poppo, Laura, and Todd R. Zenger. 1998. "Testing Alternative Theories of the Firm: Transaction Cost, Knowledge-Based, and Measurement Explanations for Make-or-Buy Decisions in Information Services." *Journal of Strategic Management* 19 (9): 853–877.
- Powell, Walter W., Kenneth W. Koput, and Laurel Smith-Doerr. 1996. "Interorganizational Collaboration and the Locus of Innovation: Networks of Learning in Biotechnology." *Administrative Science Quarterly* 41(1): 116–145.
- Prahalad, C. K., and Gary Hamel. 1990. "The Core Competence of the Corporation." *Harvard Business Review* 68 (3): 79–91.
- Quélin, Bertrand. 2002. *Les frontières de la firme*. Paris: Economica.
- Shelanski, Howard A., and Peter G. Klein. 1995. "Empirical research in Transaction Cost Economics. A Review and Assessment." *Journal of Law, Economics, and Organization* 11 (2): 335–361.
- Shimizu, Koichi. 1999. *Le Toyotisme*. Paris: La Découverte.
- Scholz, Trebor. 2016. *Platform Cooperativism*. New York: Rosa Luxemburg Stiftung.
- Tapscott, Don, and Anthony D. Williams. 2010. *Macrowikinomics: Rebooting Business and the World*. New York: Portfolio Hardcover.
- Teece, David. 2006. "Reflections on 'Profiting from Innovation.'" *Research Policy* 35 (8): 1131–1146.
- Teece, David. 2010. "Business Models, Business Strategy and Innovation." *Long Range Planning* 43: 172–194.
- Veblen, Thorstein. 1923. *Absentee Ownership. Business Enterprise in Recent Times: The Case of America*. New Brunswick: Transaction Publishers.
- Williamson, Oliver E. 1975. *Markets and Hierarchies*. New York: The Free Press.
- Williamson, Oliver E. 1980. "The Organization of Work." *Journal of Economic Behavior and Organization* 1 (1): 5–38.
- Williamson, Oliver E. 1985. *The Economic Institutions of Capitalism*. New York: The Free Press.
- Williamson, Oliver E. 1986. "The Vertical Integration of Production: Market Failure Considerations." In *Economic Organization: Firms, Market and Policy Control*, edited by Oliver Williamson, 85–100. New York: New York University Press.
- Williamson, Oliver E. 1991. "Comparative Economic Organization: The Analysis of discrete Structural Alternatives." *Administrative Science Quarterly* 36: 269–296.
- Williamson, Oliver E. 1996. *The Mechanisms of Governance*. New York: Oxford University Press.
- Williamson, Oliver E. 1999. "Strategy Research: Governance and Competence Perspectives." *Strategic Management Journal* 20: 1087–1108.
- Williamson, Oliver E. 2000. "The New Institutional Economics: Taking Stock, Looking Ahead." *Journal of Economic Literature* 38: 595–613.

Copyright of Journal of Economic Issues (Taylor & Francis Ltd) is the property of Taylor & Francis Ltd and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.